Contact & Info

Email: mialealexander@gmail.com

Links: GitHub, LinkedIn

Location: Boston, Massachusetts

Education

School: Northeastern University

Degree: BS in Data Science, Minor in Music Technology

GPA: 3.8/4.0, magna cum laude

Graduated: April 2026

Relevant Courses: Computer Science Foundations I and II, Machine Learning I and II, Artificial Intelligence, Practical Neural Networks, Information Presentation and Visualization, Human/Computer Interaction, Large-Scale Storage/Retrival, Calculus I and II, Probability and Statistics, Linear Algebra

Technical Skills

Langauges: Java, Python, R, C++

Databases: SQL, Redis, Neo4j, MongoDB, Splunk

Packages and Libraries: NumPy, Pandas, scikit-learn, Matplotlib, TensorFlow, Tensorboard, Torch, Git, Docker, Tableau, ggplot, rdbms

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Work Experience

Research Assistant

@ Northeastern University (August 2025 - April 2026)

- Conducted research with Professor Victor Zappi to **develop modern machine learning solutions** for **musical AI models** that run on **accelerated processors** to be used in **real-time** scenarios
- Experimented with both **grey box** (differentiable digital signal processing) and **black box** (neural networks) solutions
- Built **two audio effect plugins** for digital audio workstations that emulate a **TS9 distortion pedal** black box model
- Built an audio effect plugin to emulate the **LA-2A levelling compressor** grey-box model
- Awarded the **PEAK Fellowship: The Summit** for the spring semester to continue the research and present findings at **RISE**

Crew Member

@ Trader Joe's (August - December 2025)

- **Upheld the values** of Trader Joe's to ensure a perfect experience for the customers
- Aided customers with the checkout processes and **created a welcoming environment** at the registers and demo stations
- Stocked shelves to **keep the store presentable** and allow customers to find all of their favorite items
- **Assisted customers** with any issues or questions they had

Technical Projects - Spring 2026

EPV and xG Models

- Created two machine learning models that **predict the likelihood of a goal** being scored given **different player position and situational characteristics**
- Utilized **SkillCorner's open data**, containing 10 matches of Australian A-League **tracking and dynamic event data**
- Each model required different pieces of data, so two **different preprocessing pipelines** were created in **Python**
- Implemented a **neural network using PyTorch** in Python for the EPV model, and a **Bayesian logistic regression Model in R** for xG
- Finished with **0.99** ROC/AUC for EPV and **0.73** ROC/AUC for xG

Resume Score Prediction

- **Analyzed** the effectiveness of **different natural language processing techniques** to predict **resume and job compatibility**
- Models consisted of linear regression baseline, deep averaging network, LSTM/Bi-LSTM, and transformer architecture
- Combined **pure text data** with **structured numerical statistics** to further **understand what data was pertinent** to the issue
- Utilized **GloVe DOLMA embeddings** for **tokenization** and relativized the embeddings to **increase processing efficiency**
- Found **increasing correlation** with each model, as expected